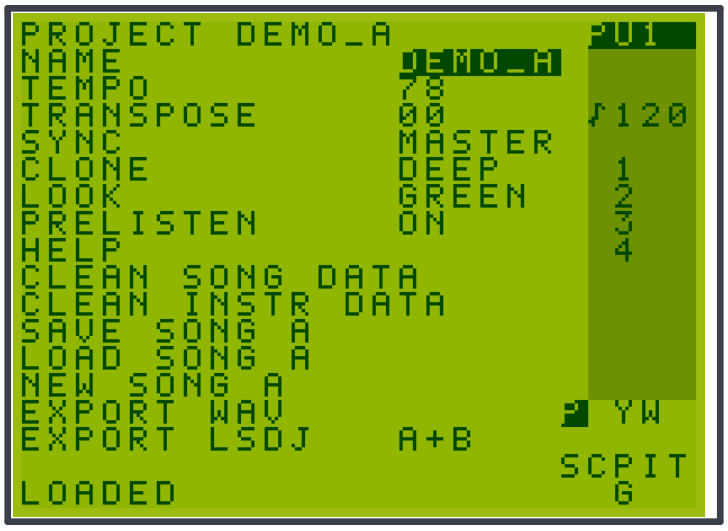
PTSDJ

The Manual



The Project page, where you save, load and export.

A Game Boy-style music maker for the Polyend Tracker

Version 1.0.0 - written for beginners. New to PTSDJ? Read the Quick Start first.

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1. Welcome

PTSDJ turns your Polyend Tracker into a music maker that works like the Game Boy program **LSDJ**. If you have never used LSDJ, that is fine. This manual starts at the beginning.

You do not need to read music. You do not need to know how to code. You will press keys, watch the screen, and listen.

What can you do with PTSDJ?

- Write songs with 8 tracks: 4 for each of two pretend Game Boys.
- Make sounds: beeps, wobbly waves, drums and noise.
- Make your own drum kits from WAV files or from a microphone or line cable.
- Save your songs on the SD card.
- Send your songs to a **real Game Boy** running LSDJ, to a flash cartridge, or to an emulator.

How music is built

LSDJ-style music is built from small parts. The small parts fit into bigger parts. It is like building with blocks.

Part	What it is	How big
Step	One beat of music. It can hold a note, a sound choice and a command.	The smallest part
Phrase	A row of 16 steps. The notes live here.	16 steps
Chain	A list of phrases that play one after another.	Up to 16 phrases
Song	A big plan. It lists which chain plays on each track.	256 rows, 4 tracks
Instrument	A sound you can play. You pick one for each step.	64 instruments
Table	A tiny pattern that changes a sound while it plays.	32 tables, 16 rows each

Here is how it works. You write a **phrase** with a few notes. You put the phrase in a **chain**. You put the chain in the **song**. Then you press **Play**.

Why is this good? You write a phrase once and use it again and again. A short drum beat can play all through a song. You only change it where you want to.

The four tracks

A Game Boy has four sound makers. PTSDJ calls them **tracks**. Each track makes a different kind of sound:

Track	Name	What it makes
1	PU1 (pulse 1)	Square-wave beeps. Good for tunes and bass.
2	PU2 (pulse 2)	Also square-wave beeps. Good for a second tune.
3	WAV (wave)	Wobbly wave sounds. Also plays drum kits (short recorded sounds).
4	NOI (noise)	Hissing noise. Good for hi-hats and crashes.

Which track plays a sound depends on the **instrument type**. A pulse instrument plays on PU1 or PU2. A wave or kit instrument plays on WAV. A noise instrument plays on NOI.

Two Game Boys

PTSDJ has **two** pretend Game Boys. **Song A** is tracks 1 to 4. **Song B** is tracks 5 to 8. They can play at the same time, and you can link them so they keep the same beat. Chapter 8 tells you how.

Counting with letters (hex numbers)

Many numbers on the screen use letters. This is called **hex**. Hex counts like this:

0 1 2 3 4 5 6 7 8 9 A B C D E F and then **10 11 12 ... 19 1A 1B ... 1F 20**

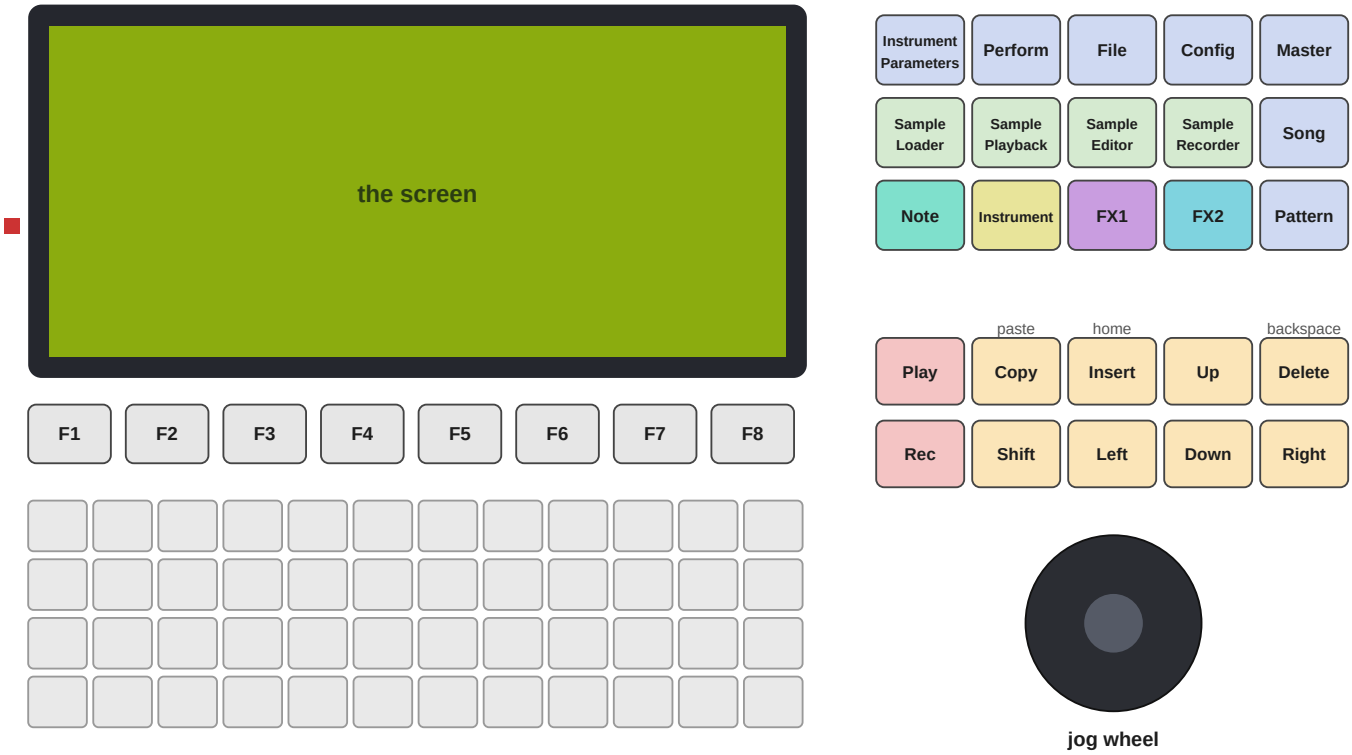
So **0A** means ten. **10** means sixteen. **FF** means 255. You do not need to do math. Just treat the letters like numbers that come after 9.

How to read this manual

- Key names look like this: **Insert**. They match the names on the Tracker.
- Hold two keys together? It says **Shift + Insert**. Hold the first and press the second.
- Page names are written like the Song page, the Phrase page and the Instrument page.
- Green boxes are tips. Red boxes are warnings. Blue boxes are good things to know.

2. Your Tracker and PTSDJ

This chapter shows the keys on your Polyend Tracker and what they do in PTSDJ. The picture matches the real Tracker.



48 pads: they play sounds, or type notes

go to a page sample keys edit and move play and record

GOOD TO KNOW

The small red light on the left side of the screen frame is the **battery light**. It glows while a song plays. In the Sample Editor it flickers when a sound is loud (Chapter 11).

The page keys

Each page key jumps straight to one page. These are the blue keys in the picture.

Key	Page it opens
Song	Song page: the big plan.
Pattern	Chain page.
Note	Phrase page: where the notes go.

Key	Page it opens
Instrument	Instrument page: your sounds. Hold **Shift** and press it for the instrument library.
Instrument Parameters	Table page.
FX1	Synth page.
FX2	Wave page.
Config	Project page: settings, saving, exporting and the help pages.
Perform	Groove page: how long each step lasts.
File	The song list for the current Game Boy.
Master	Shows both Game Boys side by side, when they are linked (Chapter 8).

The sample keys

These green keys work with drum kits. Chapter 11 explains them.

Key	Page it opens
Sample Loader	Pick a WAV file and turn it into a drum sound.
Sample Editor	Cut a sound, name it, and add it to a kit.
Sample Playback	The kit page: hear and change the sounds in a kit.
Sample Recorder	Record a sound from a line cable or a microphone.

The editing keys

These are the keys you press most. The names in this manual match the Game Boy buttons you may know.

Key	Like on a Game Boy	What it does
Insert	A button	Puts something in, or does the main thing.
Delete	B button	Clears or goes back. The key says Backspace above and Delete below. This manual calls it **Delete** or BKSP.
Shift	Select	Hold it to do more. **Shift** + arrows move between pages.
Play	Start	Starts and stops the music.
Arrows	D-pad	Move the bright cursor box.
Copy		Copies. Hold **Shift** and press **Copy** to paste. The key shows Paste above its name.

Key	Like on a Game Boy	What it does
Rec		Turns on recording from the pads (Chapter 7).
Jog wheel		Turn it to change a number. Same as holding **Insert** and pressing Left or Right.

The pads

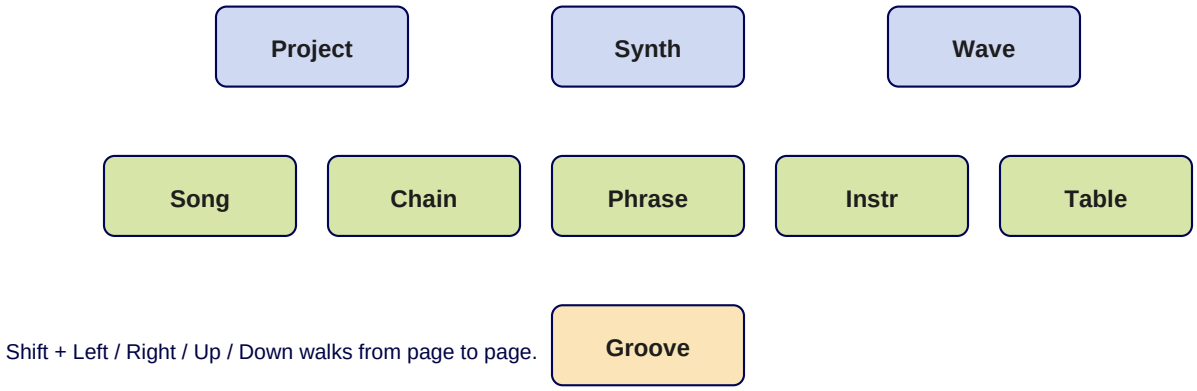
The 48 pads are at the bottom left. Press a pad and you hear the sound that is on the screen. On the Phrase page they can also type notes for you. Hold **Shift** and press a pad row to change the octave (how high the pads play).

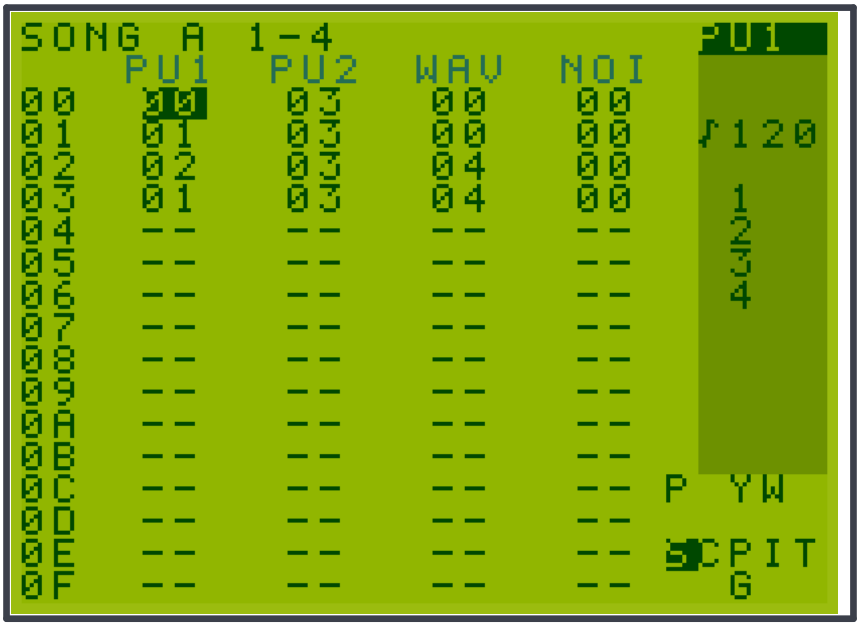
The F keys

The eight keys under the screen are F1 to F8. Hold F8 to see a help box about the page you are on. F1 and F2 turn the pages of that help box. When two songs are shown side by side, the F keys also set volume and muting (Chapter 8).

The page map

The pages sit next to each other, like rooms in a house. Hold **Shift** and press an arrow to walk to the next one. You can also use the page keys to jump.





The Song page. The panel on the right shows the track name, the tempo (the number after the note symbol), the track numbers 1 to 4, and the page map. The bright letter is the page you are on.

3. Setting up

The SD card

PTSDJ keeps everything on the SD card. These are the folders it uses. It makes most of them by itself the first time they are needed.

Folder or file	What it is for
GB/LSDJ.GB	Your own copy of the LSDJ ROM. **You must supply this.** PTSDJ needs it for drum kits, the synth page, and to build ROMs for your Game Boy.
GB/Saves/A.srm, B.srm	Your songs. Song A goes in A.srm and song B in B.srm. Each file holds up to 32 songs.
GB/Songs	Loose LSDJ song files (.lsdsng) that you can load.
GB/Instruments	Saved sounds (.gbi). The presets go here too.
GB/Samples	WAV files that you turn into drum sounds.
GB/Kits/A and GB/Kits/B	Your drum kits. Each Game Boy has its own kits.
GB/Export	What PTSDJ writes when you export: WAV files, save files and ROMs for a real Game Boy.
Firmware	Update files (.ptf) for the Tracker, and the small ot_restore.txt file.

CAREFUL

PTSDJ does not include the LSDJ ROM. Use your own copy. Songs, kits and presets are made by you or by PTSDJ. The ROM is not.

Installing PTSDJ

1. Put the **.ptf** file for PTSDJ in the **Firmware** folder on the SD card.
2. Put **ot_restore.txt** in the same folder. It has one line: the exact file name of the .ptf. Nothing else. No extra lines.
3. Put the card in the Tracker. Turn the Tracker off and on.
4. The Tracker installs the new firmware and restarts. PTSDJ starts with the letters PTSDJ dropping down and the Game Boy chime.

TIP

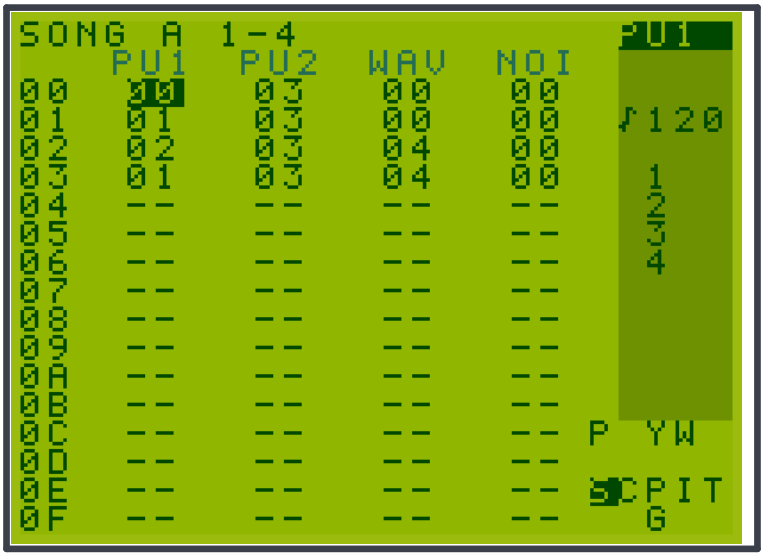
If PTSDJ is already installed you can also update from inside it. Open the Project page, hold **Shift**, and press F8. The ? changes to FIRMWARE. Pick a file and press **Insert** twice. Chapter 13 has the details.

Going back to the normal Tracker

Put the stock Polyend .ptf file in the Firmware folder and use the same steps. Your PTSDJ files stay on the SD card. Normal Tracker projects are never touched by PTSDJ.

The first start

The first time, there is no saved song, so both Game Boys open with a **demo song**. DEMO_A is a melody. DEMO_B is a beat. Press **Play** to hear them together. After that, each Game Boy opens the song you saved last. The first screen is the Song page.



The Song page of DEMO_A. The numbers are chains. A -- is an empty spot.

To start a song of your own, press the **File** key and pick **NEW SONG**. Chapter 9 shows how.

4. Editing basics

Every page works in the same way. Learn these ideas once and you know how to use every page.

The cursor

The cursor is the bright box. Press the arrows to move it. It shows where **Insert** and **Delete** will work.

Putting things in

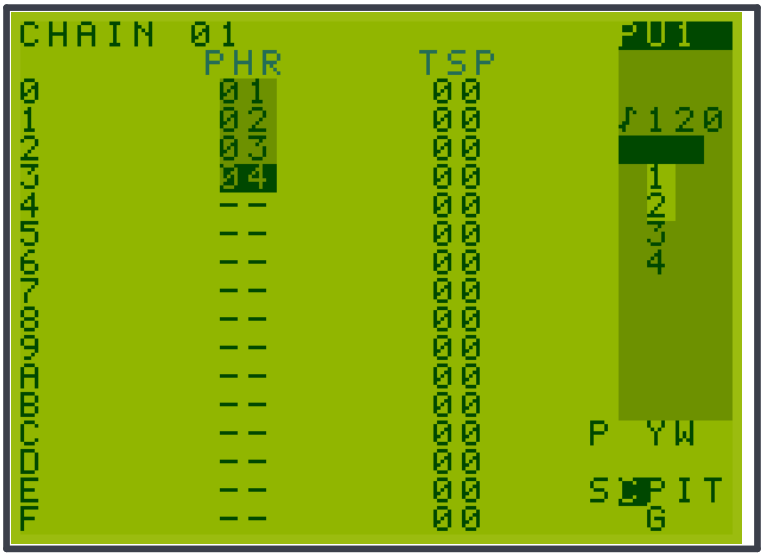
1. Move the cursor to a box.
2. Press **Insert**. The box fills with a value. It is usually the last value you used.
3. To change it, turn the jog wheel. Or hold **Insert** and press an arrow.

Do this	What happens
Insert	Puts in a value.
Insert + Left or Right	Changes the value by 1.
Insert + Up or Down	Changes the value by 16 (or by an octave on notes).
Jog wheel	Same as **Insert** + Left or Right.
Delete	Clears the box.
Delete + Left or Right	On Chain, Phrase, Instrument, Table, Groove and Wave pages: choose another chain, phrase, and so on.

Picking many boxes at once

Hold **Shift** and **Insert** together, then press the arrows. The boxes you cover light up. Now:

- **Delete** clears all of them.
- The jog wheel changes all the filled ones at once.
- **Copy** copies them. Move somewhere else, then hold **Shift** and press **Copy** to paste.



Three boxes picked on the Chain page.

GOOD TO KNOW

Copy and paste works on the Song, Chain, Phrase, Table and Groove pages, and only inside the same page. If you paste a chain or phrase, the Project page setting CLONE decides what happens. **DEEP** makes brand new copies of the phrases. **SLIM** reuses the same ones.

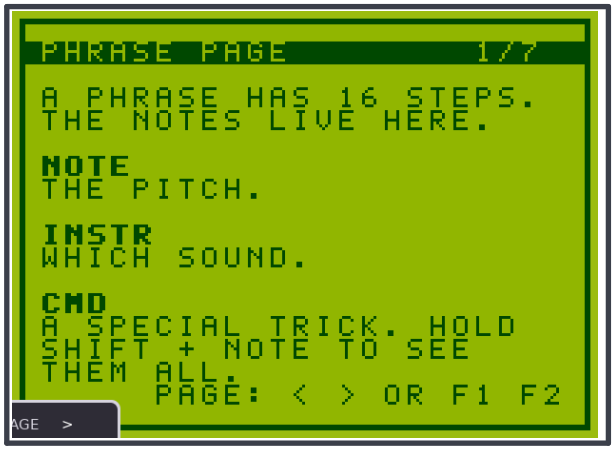
Moving between pages

You can jump with a page key, or walk with **Shift** and the arrows. When you walk right from the Song page, PTSDJ follows what you were on. If the cursor was on chain 03, the Chain page opens at chain 03. If you were on a phrase, the Phrase page opens at that phrase. It is a fast way to dive into your song.

Getting help on the Tracker

PTSDJ has help built in. You do not need this manual to look up a key.

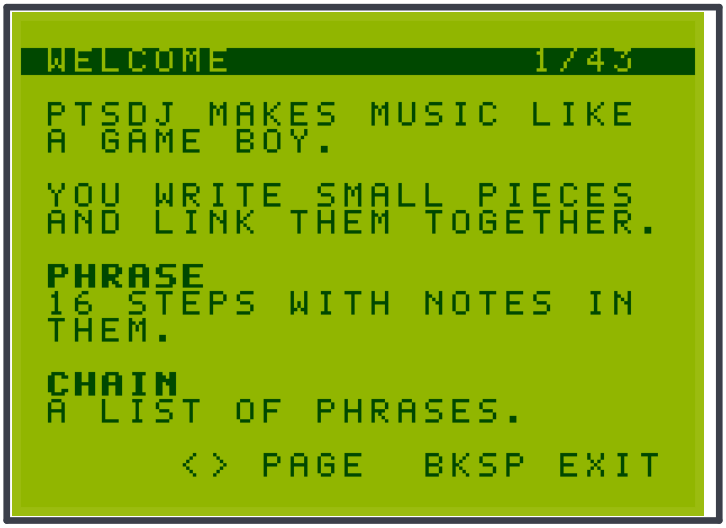
- **Hold F8.** A box pops up about the page you are on. Press Left or Right to turn its pages. Or use F1 (back) and F2 (forward) with your other hand. A little tab with < and > grows out of the frame above them.
- **The cheat sheets.** On the Phrase page, hold **Shift** and press **Note** to see every command. On the Table page, hold **Shift** and press **Instrument Parameters**.
- **The help pages.** Open the Project page and move to HELP. Press **Insert**. Left and Right turn the pages. **Delete** leaves.



Hold F8: a pop-up about the Phrase page.



The command cheat sheet.



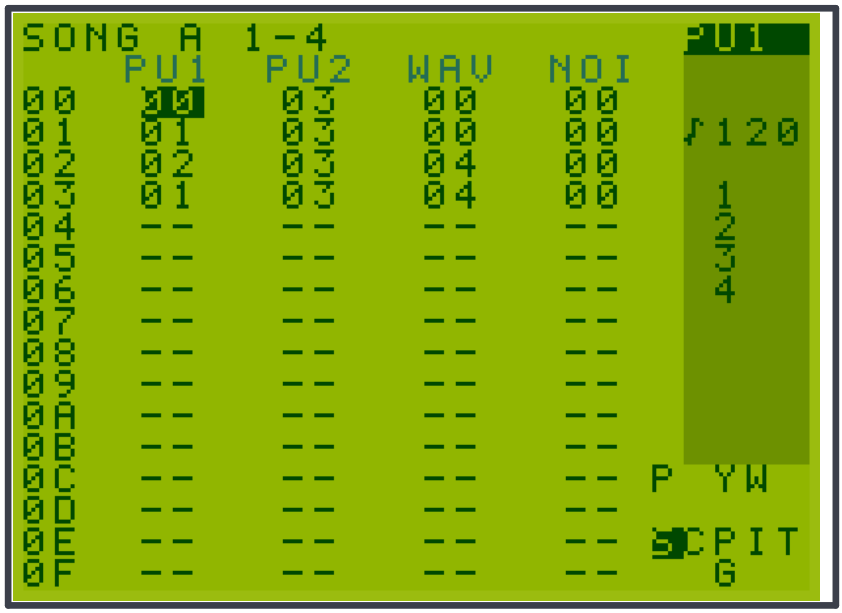
The help pages start with a welcome.

5. The pages, one by one

This chapter walks through every page. For each one you see what it is for, what is on the screen, and which keys to press.

The Song page

The Song page is the big plan. Each row has four boxes, one for each track: PU1, PU2, WAV and NOI. A box holds a **chain number**. A box with -- is empty. All four tracks play down the rows at the same time.



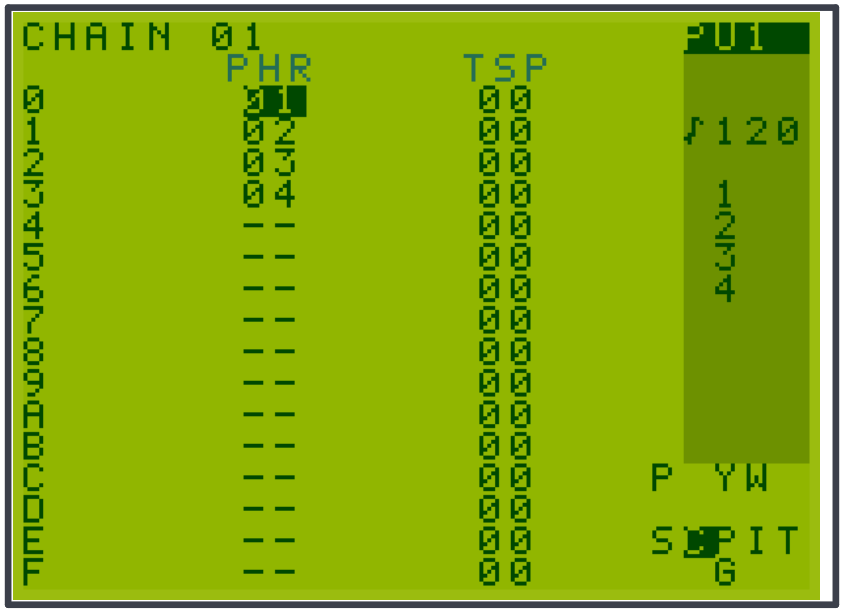
The Song page of a finished song. The numbers are chain numbers.

Key	What it does
Insert	Puts a chain number in the box. Then hold **Shift** and press Right to open that chain.
Delete	Clears the box.
Play	Starts the song at the row you are on. Press it again to stop.
Shift + **Insert**	Mutes that track (and unmutes it).
Shift + **Delete**	Mutes all the tracks of this Game Boy.
Shift + **Song**	Switches between song A and song B.
Shift + **Insert** + arrows	Picks many boxes. **Copy** and **Shift** + **Copy** copy and paste them.

> **TIP**
>
> When a track runs out of chains it starts again at the top of its column. You can make a long song with short columns that repeat.

The Chain page

A chain is a list of up to 16 phrases. They play one after another. The left column is the **phrase** number. The right column is **TSP**, which means transpose: it moves all the notes of that phrase higher or lower. **0C** is one octave up.



A chain with four phrases. The 03 phrase plays four times.

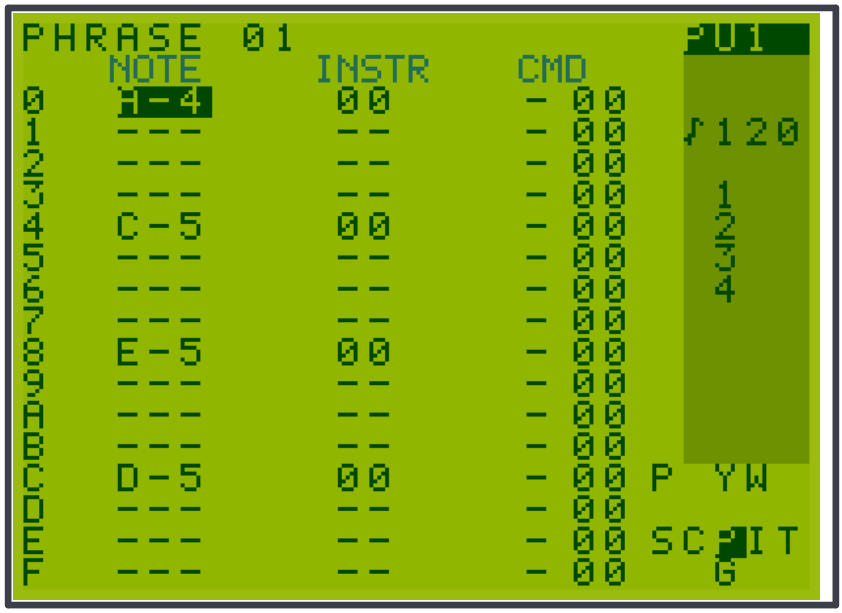
Key	What it does
Insert	Puts a phrase number in the box. Hold **Shift** and press Right to open the phrase.
Delete + Left or Right	Picks another chain.
Play	Plays just this chain and loops it. **Shift** + **Play** plays the whole song from here.
Left column / right column	Move between the PHR box and the TSP box with Left and Right.

The Phrase page

A phrase has 16 steps, numbered 0 to F. Each step has three boxes:

- **NOTE** is the pitch. It looks like C-3 or F#4. C-3 is a lower C and C-4 is a higher one.
- **INSTR** is which sound plays. It is an instrument number.

- **CMD** is a special trick. It is a letter and a two-digit number. See Chapter 6.



A phrase. Notes come with the instrument number and a command.

Key	What it does
Insert	Puts in a note. The first note you put in uses the last instrument you picked.
Insert + Left or Right	Changes the note by one half step.
Insert + Up or Down	Changes the note by one octave (12 half steps).
Delete	Clears the step.
Delete + Up or Down	Goes to another phrase.
Play	Plays just this phrase on its track and loops it. **Shift** + **Play** plays the whole song.
Pads	With **Rec** off, a pad types its note at the cursor. With **Rec** on, the pads only make sound (see Chapter 7).

TIP

There is no OFF note in LSDJ, and none in PTSDJ. To end a note, put the **K** command on the step. K08 ends the note after 8 ticks (a step is 6 ticks). The notes run from C-2 up to a top note. A big step with Up or Down that would go past the end does nothing.

GOOD TO KNOW

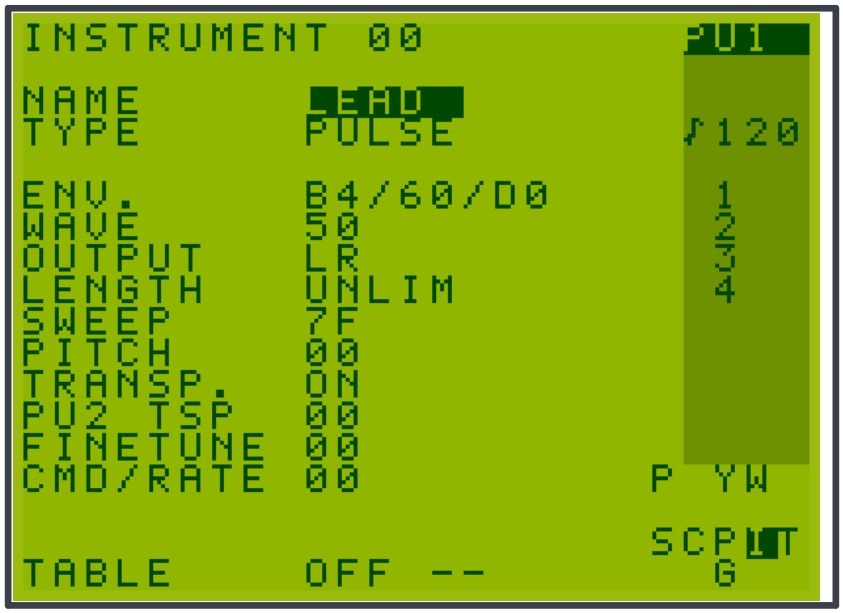
For a drum kit sound, the NOTE column does not show a pitch. It shows the **name of a drum sample**. Chapter 11 explains drum kits.

The Instrument page

An instrument is a sound. PTSDJ has 64 of them, numbered 00 to 3F. The number is at the top of the page. To pick a different one, move the cursor up onto the number and use Left and Right or the jog wheel. The pads play the sound as you go.

The first row is **NAME**. Press **Insert** to type a name of up to five letters. The second row is **TYPE**. It can be PULSE, WAVE, KIT or NOISE. The rows below change with the type.

> **GOOD TO KNOW**
>
> A new instrument starts as a plain pulse sound you can hear right away. When you change the TYPE, the sound goes back to the plain settings of that type.



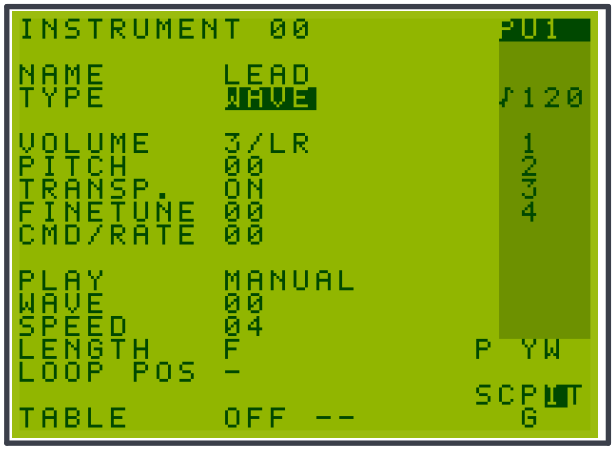
A pulse instrument.

Rows with more than one value

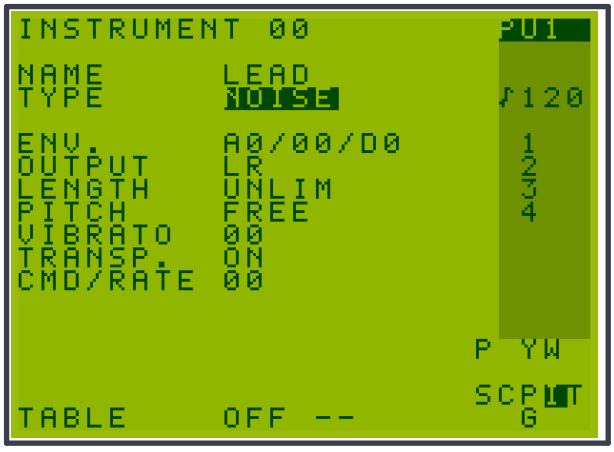
Some rows hold two or three values side by side, like **ENV.** (three) and **TABLE** (two). The bright box shows which one you are on. Press Left or Right to move to the next value. Then use **Insert** with Left or Right, or the jog wheel, to change it.

Pulse instruments (tracks PU1 and PU2)

Row	What it does
ENV.	How loud the note is and how it fades. It has three two-digit values. In the first, the first digit is how loud the note starts (0 is silent, F is loudest) and the second is how fast it fades (0 means no fade; small numbers fade fast). The second and third values add two more stages, so a note can start loud, drop to a middle level, then fade out.
WAVE	The shape of the beep: 12.5, 25, 50 or 75. Thin sounds (12.5) are buzzy. 50 is round and full.
OUTPUT	Which speaker: OFF, L (left), R (right) or LR (both).
LENGTH	UNLIM lets the note ring until the next one. A number makes the note stop by itself. Small numbers are short.
SWEEP	Makes the pitch slide at the start of each note. 7F means no sweep. Only works on PU1.
PITCH	The style of pitch bend, slide and vibrato. The usual value is fine to start.
TRANSP.	ON: the note follows the transposes of the chain and the project. OFF: it ignores them.
PU2 TSP	Moves the pitch of PU2 up or down in half steps. 0C is one octave.
FINETUNE	A tiny pitch nudge. Two pulse tracks with a tiny nudge sound thick and wide.
CMD/RATE	A setting LSDJ keeps for each instrument. PTSDJ stores it so your songs stay the same on a real Game Boy, but it does not use it yet. Leave it as it is.
TABLE	Two values. The first is OFF, ON or STEP. ON runs the table as time passes. STEP moves the table one row for each new note. The second is the table number (Table page). While the table is OFF you cannot change the number.



A wave instrument.



A noise instrument.

Wave instruments (track WAV)

A wave instrument plays a little repeating shape that you design on the Synth and Wave pages.

Row	What it does
VOLUME / OUTPUT	Two values. VOLUME is 3 (full), 2 (half), 1 (a quarter) or 0 (silent). Next to it is the speaker choice.
PITCH, TRANSP., CMD/RATE, TABLE	Work like on the pulse instrument.
FINETUNE	A tiny pitch nudge.
PLAY	MANUAL plays one wave all the time. ONCE, LOOP and PINGPONG step through the 16 waves of a synth by themselves: once, round and round, or back and forth.
WAVE (or SYNTH)	Which wave to play. With PLAY on MANUAL it is a two-digit wave number: the first digit is the synth (0 to F) and the second is which of its 16 waves. With the other PLAY settings the row is called SYNTH and you pick only the synth, 0 to F. When you turn PLAY up from MANUAL, the wave number goes back to the first wave of its synth.
SPEED	How many ticks each wave stays, when PLAY steps through them. Small numbers are fast.
LENGTH	How many of the 16 waves are used when PLAY steps through them. F uses all 16.
LOOP POS	Where a LOOP starts again. A dash means from the first wave. It cannot go before the first wave that is used.

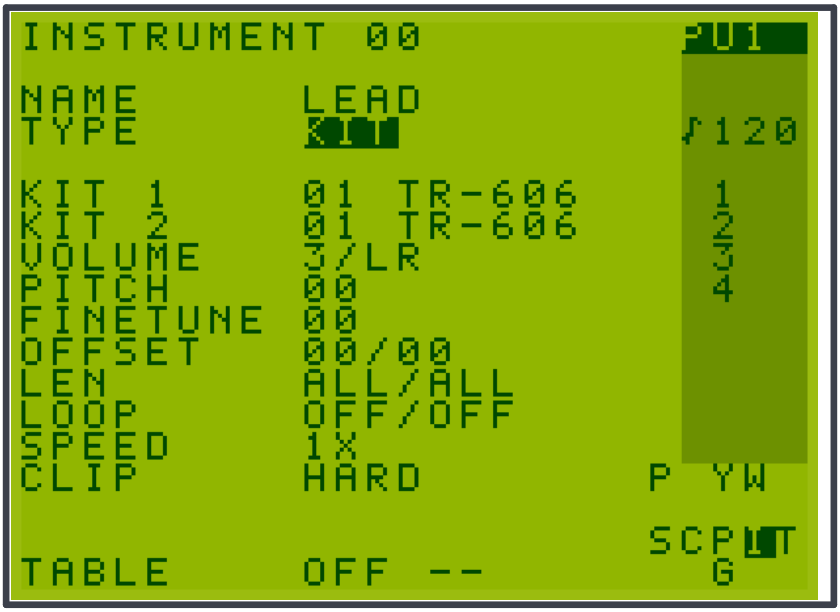
> **TIP**
>
> A wave note has no fade. It rings until the next note. To end it, put the **K** command on the step (Chapter 6). The same goes for the end of a song: if a song loops back to a quiet part, end the last wave note with K or it will ring all through the quiet part.

Noise instruments (track NOI)

Row	What it does
ENV.	Loudness and fade, like the pulse instrument.
OUTPUT, LENGTH, TRANSP., CMD/RATE, TABLE	Work like on the pulse instrument.
PITCH	FREE or SAFE. These are the same two choices LSDJ has for how the noise pitch is picked.
VIBRATO	The style of vibrato for the noise. The usual value is fine to start.

Kit instruments (track WAV)

A kit instrument plays short recorded sounds, such as drums. It plays on the WAV track. Chapter 11 has the full story.



A kit instrument. KIT 1 and KIT 2 name the drum kits. The kit name shows next to each number.

Row	What it does
KIT 1	The first drum kit (01 to 33 in hex, counting from 01 like a real Game Boy). The step's note picks a sample from this kit.
KIT 2	A second kit. It plays at the same time as KIT 1 when a step picks a KIT 2 sample.
VOLUME / OUTPUT	Two values. VOLUME is 3 (full), 2 (half), 1 (a quarter) or 0 (silent). Next to it is the speaker choice.
PITCH, TABLE	Work like on the other instruments.
FINETUNE	A tiny pitch nudge for every sample.
OFFSET	Where in the sample the sound starts. Bigger numbers skip more of the beginning. The second value is the one PTSDJ uses.
LEN	How much of the sample plays. 00 (shown ALL) plays all of it. A number cuts it short. The second value is the one PTSDJ uses.
LOOP	Two ON / OFF values. The second one repeats the sample. PTSDJ uses only that one.
SPEED	1X is the normal speed. 0.5X plays the sample at half speed, so it sounds an octave lower and lasts twice as long.
CLIP	What happens when two layered samples add up to too much: HARD, SOFT, FOLD or WRAP. PTSDJ always clips hard, like HARD. The setting is kept for a real Game Boy.

Naming an instrument

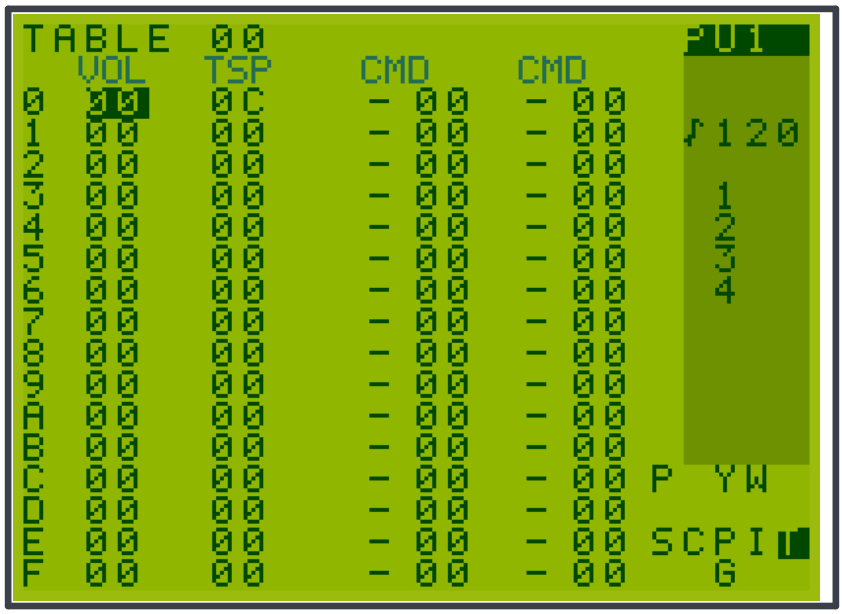
On the NAME row, press **Insert**. A little keyboard opens. Move with the arrows. Press **Insert** to type the letter. The < key erases. Press **Play** for OK. Press **Delete** to cancel.



The name keyboard.

The Table page

A table is a tiny pattern that plays by itself while a sound plays. It has 16 rows. Each row can change the volume, change the pitch, or run a command. There are 32 tables (00 to 1F).



A table with commands in it.

Column	What it does
VOL	Loudness for that row.

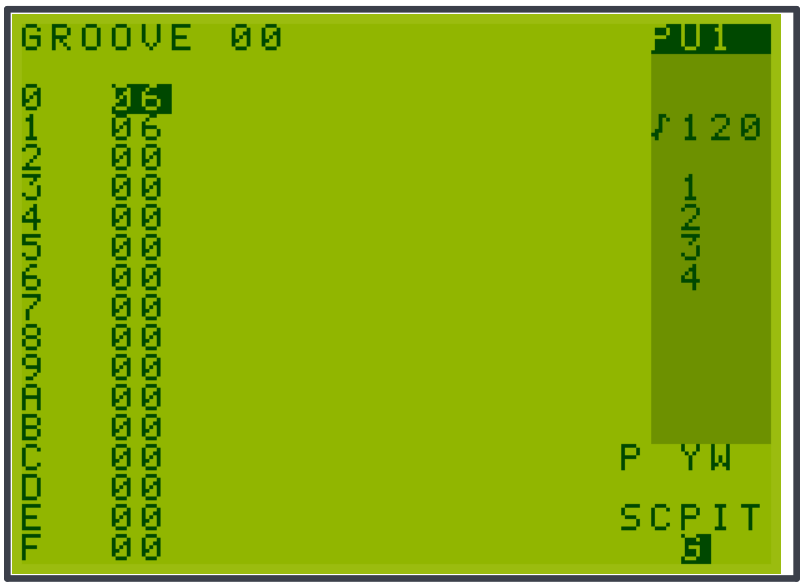
Column	What it does
TSP	Pitch change in half steps. 0C moves the pitch up by one octave.
CMD and value (two of them)	Two commands per row. Hold **Shift** and press **Instrument Parameters** to see all of them.

To use a table: go to the Instrument page, move to TABLE, and pick the table number. The table now plays every time that instrument plays. The table moves one row at a time and loops back to the top after row F. A hop command (H) inside a table can jump to another row instead.

Press **Play** on the Table page to hear the sound that uses it. Press **Play** again to stop. **Shift** + **Play** plays the whole song.

The Groove page

A step lasts a number of **ticks**. A tick is one small tap of the clock. The groove says how many ticks each step lasts. The default is **06**, so every step lasts the same time.



The Groove page. Step 0 and step 1 both last 6 ticks.

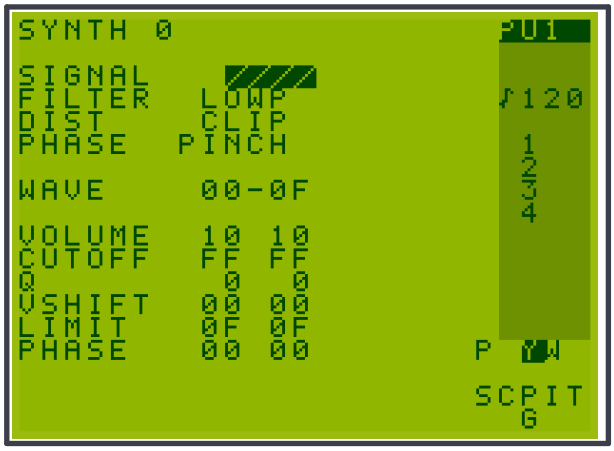
Groove	Sounds like
06 06	An even beat.
07 05	Swing: long step, short step.
04 08	A stronger bounce.

The list repeats. A **00** ends the list at that step. There are 32 grooves (00 to 1F). In a phrase, the command **G** picks a groove. **Delete** clears a step to 00. **Delete** + Left or Right picks another groove.

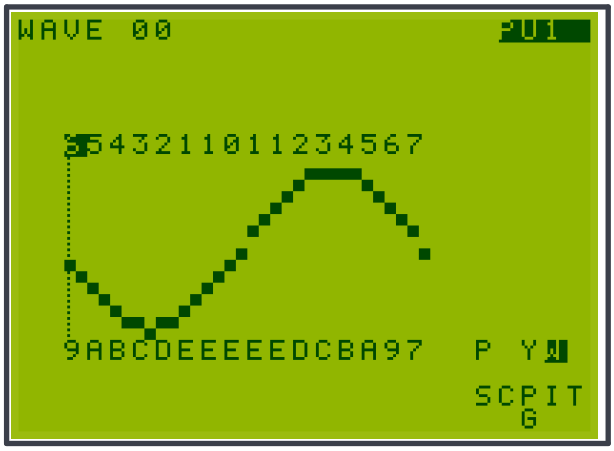
The Synth page and the Wave page

The WAV track makes its sound from a 32-dot shape called a **wave**. The Wave page lets you draw one. The Synth page makes waves for you with a few knobs.

> **CAREFUL**
>
> The Synth page needs **GB/LSDJ.GB** on the SD card. It uses the ROM to build the waves.



The Synth page. Each row shapes the sound: filter, distortion, phase and volume.



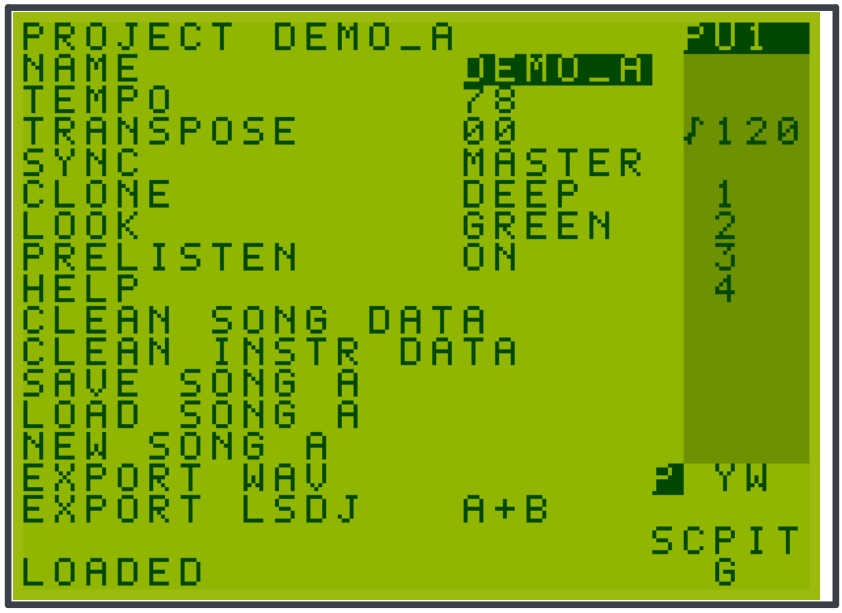
The Wave page. Each dot is one of 32 points in the shape.

On the Synth page	What it does
Left or Right	Moves to another digit.
Insert + arrows	Changes a value.
Play	Plays the synth so you can hear it.
Delete + Left or Right	Picks another synth (0 to F).

On the Wave page	What it does
Left or Right	Moves along the dots.
Up or Down	Moves one dot higher or lower.
Play	Plays the wave.
Delete + Left or Right	Picks another wave.
Delete (a quick tap)	Copies the wave.
Shift + **Insert**	Pastes the copied wave.

The Project page

This page holds the settings for the song on the current Game Boy. Press **Config** to open it. It also has the buttons to save, load and export.



The Project page.

Row	What it does
NAME	The name of the song. Press **Insert** to type it.
TEMPO	How fast the song plays. Bigger is faster. It is a hex number from 28 to FF. 80 is a medium speed.
TRANSPOSE	Moves every note up or down in half steps.
SYNC	OFF, MASTER or SLAVE. Used when two Game Boys play together (Chapter 8).
CLONE	DEEP or SLIM. Decides what happens when you paste chains and phrases (Chapter 4).
LOOK	The colours of the screen. Turn the jog wheel to try them.
PRELISTEN	ON: the pads make sound when you press them on any page. OFF: the pads stay quiet, except while recording.
HELP	Opens the help pages.
CLEAN SONG DATA	Frees chains and phrases that no song uses. Press **Insert** twice to say yes.
CLEAN INSTR DATA	Does the same for sounds nobody uses. Press **Insert** twice.
SAVE SONG	Saves the song to this Game Boy's save file.
LOAD SONG	Shows the songs in the save file.
NEW SONG	Asks for a name and starts an empty song.
EXPORT WAV	Makes an audio file of your song.
EXPORT LSDJ	Pick A, B or A+B with the jog wheel, then press **Insert**. Makes files for a real Game Boy (Chapter 12).

TIP

Hold **Shift** on the Project page and two things change. The ? above F8 turns into FIRMWARE, and a line of tiny technical numbers appears at the bottom. The line is for fixing problems.

6. Commands

A command is a special trick for one step. You put it in the CMD box of a phrase or a table. A command is a **letter** and two digits, like **K03**.

Two letters in the digits show up a lot: **x** and **y**. They are two separate digits (0 to F). **xx** means both digits together as one number (00 to FF).

> **TIP**
>
> Forget one? Hold **Shift** and press **Note** on the Phrase page. A cheat sheet shows every command.

Every command

Command	What it does	Example
A xx	Start table xx. A20 stops the table.	A03 starts table 03.
B xy	Maybe play the note. A big number means the note plays often. A small number means seldom.	BFF always plays. B40 plays about one time in four.
C xy	Chord: plays the note, then +x and +y half steps, one after another, very fast.	C47 makes a major chord sound.
D xx	Delays the note by xx ticks.	D03 starts the note 3 ticks late.
E xy	Sets volume x, then the fade speed y.	EA2 starts loud and fades.
F xy	Fine tune for pulse sounds.	
G xx	Picks groove number xx.	G01 uses groove 01.
H xx	Hop: jumps to step x of the next phrase. HFF stops everything.	HFF stops the song.
K xx	Kill: stops the note after xx ticks.	K02 gives a very short blip.
L xx	Slide to the new note in xx ticks.	L08 glides over 8 ticks.
M xy	Master volume. x is the left side, y is the right side (0 to 7).	M77 is full volume.
O xx	Output. 0 is off, 1 is left, 2 is right, 3 is both.	O01 plays only on the left.
P xx	Pitch bend. Goes up for small numbers. FF bends down.	
R xy	Repeat (retrigger) the note every y ticks.	R02 makes a fast stutter.
S xy	Sweep on pulse 1, or shape on noise.	
T xx	Sets the tempo.	T80 sets a medium tempo.
V xy	Vibrato: x is the speed, y is the depth.	V44 is a gentle wobble.
W xx	Pulse width (the shape of the beep).	
Z xy	Randomizes the last command.	

GOOD TO KNOW

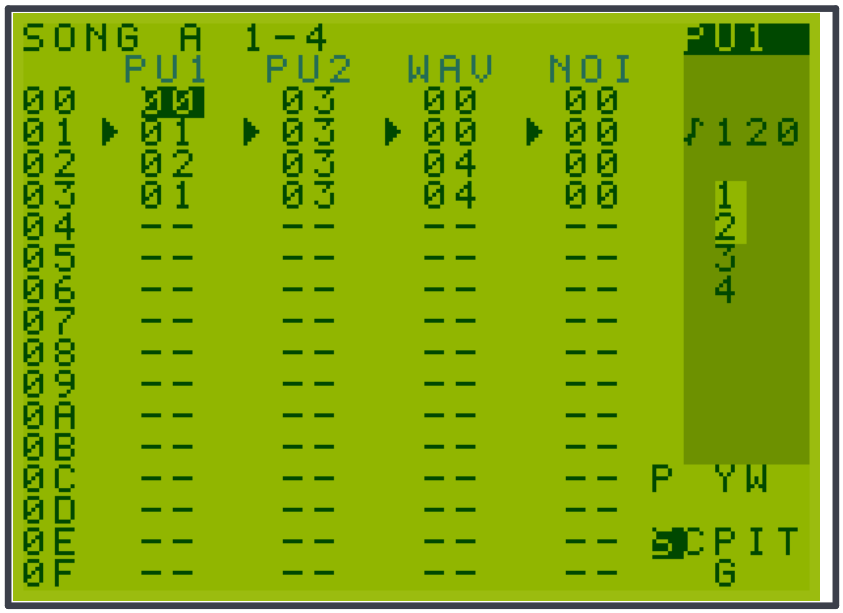
Some commands (like B and Z) use random numbers. They do not play exactly the same each time. That is part of the fun.

7. Playing and recording

Play

Play starts and stops the music. What it plays depends on the page you are on:

Page	Press **Play** to hear...	**Shift** + **Play**
Song	The song, from the row the cursor is on.	
Chain	Just this chain, looping, on the track you came from.	The whole song from here.
Phrase	Just this phrase, looping, on its track.	The whole song from here.
Table	A note using the sound that uses this table. Press Play again to stop.	The whole song.
Synth, Wave	The shape you are working on.	
Sample pages	The sample you are on (Chapter 11).	
Library	A preview of the highlighted sound (Chapter 10).	



A song playing. The small arrows next to the rows show where each track is.

Muting and soloing tracks

On the Song page, hold **Shift** and press **Insert** to mute the track under the cursor. Press it again to bring it back. **Shift** + **Delete** mutes all four tracks. This is a great way to hear one part at a time.

Tempo and swing

Set the tempo on the Project page, or with the T command. Swing comes from the Groove page: make the first step a little longer than the second, such as 07 and 05.

The pads

The pads play the sound that is on the screen. The instrument you are on decides the track: pulse on PU1 or PU2, wave and kit on WAV, noise on NOI. The pads never start the song.

- Hold **Shift** and press a pad row to change the octave. The side panel shows O+1, O-1 and so on.
- On a kit sound, pad 1 plays sample 1, pad 2 plays sample 2, and so on.

Writing notes with the pads

There are three ways to use the pads on the Phrase page. **Rec** decides which one you get.

Mode	What the pads do
Rec off	Step mode. Press a pad and its note is typed at the cursor. The cursor moves down for the next one.
Rec on	The pads only make sound. Turn the jog wheel to pick the sound. Nothing is written.
Rec on, then **Play**	Live recording. The phrase loops on its track while the whole song plays behind it. The pads write the notes you play, placed on the nearest step. Press **Play** to stop.

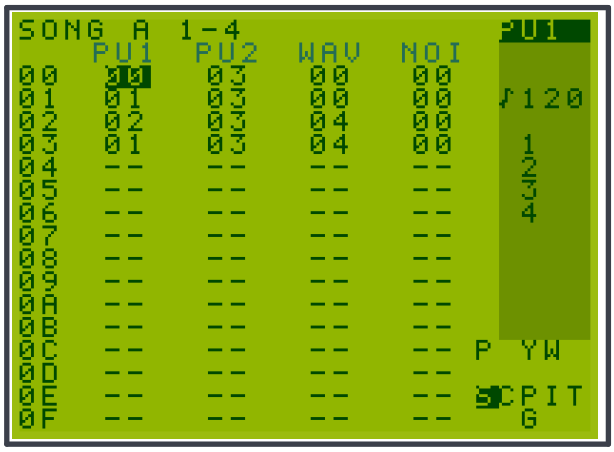
> **TIP**
>
> Recording lines up with what you hear. PTSDJ takes the delay between pressing and hearing into account.

8. Two Game Boys at once

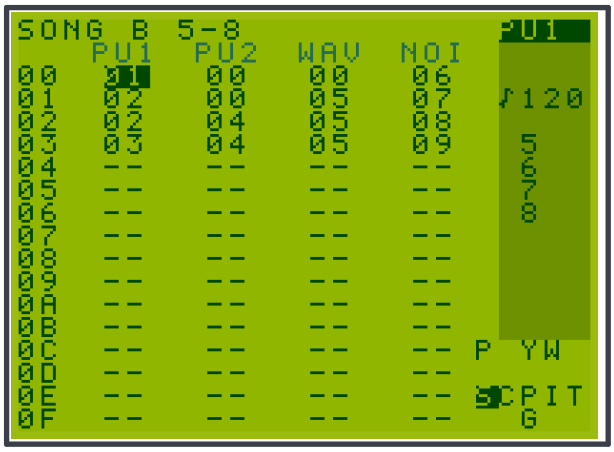
PTSDJ has two pretend Game Boys. Each has its own song, its own tempo, and its own drum kits. Together they give you 8 tracks.

Which one am I on?

The title of the Song page tells you. **SONG A 1-4** is the first Game Boy. **SONG B 5-8** is the second. Hold **Shift** and press **Song** to swap.



Song A: tracks 1 to 4.

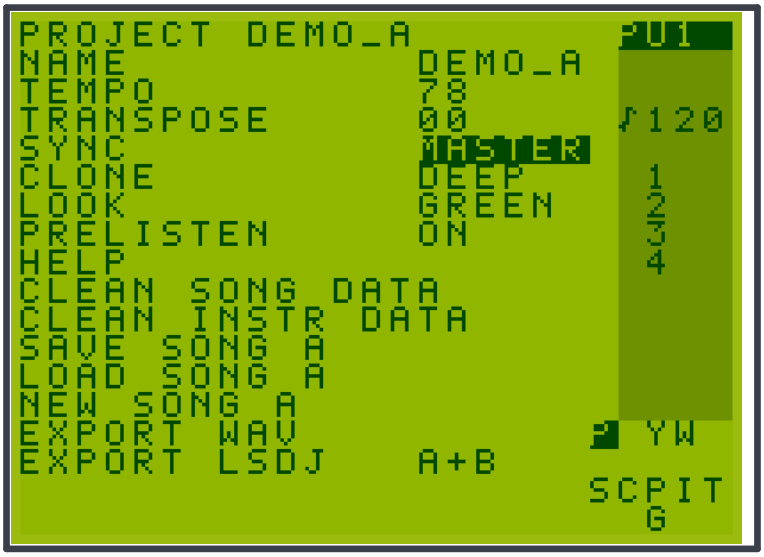


Song B: tracks 5 to 8.

Starting both together

When both songs have **SYNC** set, **Play** on either one starts both. Do it like this:

1. On song A, open the Project page. Set SYNC to **MASTER**.
2. Hold **Shift** and press **Song** to go to song B. Open the Project page. Set SYNC to **SLAVE**.
3. Press **Play**. Both play together.



SYNC set to MASTER on song A.

The **master** sets the tempo. The **slave** follows it. If you change the master's tempo, the slave's tempo changes too.

Side-by-side view

Press **Master** while the songs are linked. Both Game Boys show next to each other on the screen. The one you are working on is bright. The other is dim.



Both Game Boys side by side.

Look at the bottom corners. A letter **A** sits above F1. A letter **B** sits above F8. Hold either key and a dock rises. It names what every F key does:



The dock. Volume and mute keys for each Game Boy.

Keys	What they do
F1 and F8	Pick song A or song B. The keys now control that one.
F2 + jog wheel	Volume of song A.
F7 + jog wheel	Volume of song B.
F3 to F6	Mute track 1 to 4 of the active Game Boy. Hold **Shift** too to play only that track.

9. Saving and loading

Each Game Boy keeps its songs in its own save file on the SD card. It works like a real LSDJ cartridge. One file holds up to **32 songs**.

File	What is in it
GB/Saves/A.srm	The songs for Game Boy A.
GB/Saves/B.srm	The songs for Game Boy B.

Save

Open the Project page. Move to **SAVE SONG** and press **Insert**. The bottom line says SAVED. The song is filed under the song's name. If you save with the same name, the old song is replaced. A new name adds a new song. If all 32 spots are full you see BANK FULL.

> **CAREFUL**
>
> Save before you start a NEW SONG or load another song. Loading replaces the song on the screen.

Load

Press **File**, or choose LOAD SONG on the Project page. The list shows the songs for this Game Boy.



The song list. NEW SONG is the first line. A + sign shows a loose file from GB/Songs.

Key	What it does
Up or Down	Pick a song.
Insert	Load the song. It replaces the song on the screen.
Shift + **Delete**	Erase the picked song from the save file. Press **Insert** to say yes.
Delete	Go back.

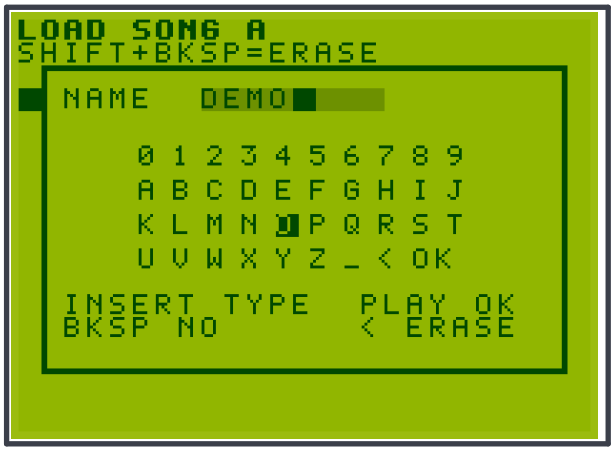
A **-** before a name means the song is in the save file. A **+** means it is a loose file in GB/Songs. A loose song moves into the save file the next time you save. A ***** means it is the built-in demo song for this Game Boy. It is always in the list and you cannot erase it.

New song

The first line of the song list is **NEW SONG**. The Project page also has a NEW SONG row. Press **Insert** and type a name. PTSDJ starts an empty song. Your kits and your saved sounds are still there.



Typing a name.



The name DEMO.

When PTSDJ starts

Each Game Boy opens the song you saved last. If neither Game Boy ever saved a song, they open with the demo songs. If only one of them saved a song, the other one opens empty.

The demo songs

DEMO_A has the melody. DEMO_B has the beat and the bass. They are made to be one piece of music: the same key (A minor), the same speed (120 beats a minute), and the same length (16 bars, about 32 seconds). SYNC is on in both, so pressing **Play** on either Game Boy starts both. Load ***DEMO_A** on Game Boy A and ***DEMO_B** on Game Boy B to bring them back after you change them. You can also use them as a starting point: change a note, save under a new name, and the original stays in the list.

Song files from LSDJ

PTSDJ can read LSDJ song files (.lsdsng). Put them in **GB/Songs**. They show up in the song list with a +. When you export, PTSDJ also writes a .lsdsng of your song.

Audio files

EXPORT WAV on the Project page makes an audio file in **GB/Export**. It plays the song once, until every track starts to repeat. You can share it, or play it on a computer.

Tidy up

CLEAN SONG DATA and **CLEAN INSTR DATA** on the Project page free up chains, phrases and instruments that nothing uses. Press **Insert** twice to confirm. This makes the song file smaller.

10. The instrument library

The library keeps your sounds on the SD card so you can use them in any song. Hold **Shift** and press **Instrument** to open it.



The library. The first line saves a sound. The others are files.

Saving a sound

1. Pick the sound you want on the Instrument page.
2. Hold **Shift** and press **Instrument**. The top row says SAVE (the sound's name) AS.
3. Press **Insert**. Type a name. Press **Play** for OK.

Using a sound

1. Press Up or Down to pick a file.
2. Press **Play** to **preview** it. Nothing in your song changes. The pads play it too.
3. Press **Insert** to put it into your song, in the instrument you were on.

Key	What it does
Up or Down	Pick a file.
Play	Preview the sound. It goes into temporary memory, so your song stays as it is.
Insert	Put it in the current instrument. This does change your song.
Left or Right	Turn a page of 13 files.

Key	What it does
Delete	Go back.

> **GOOD TO KNOW**
>
> A saved sound that uses a table brings its table along. If no empty table is free, the sound loads without it.

The presets

There are **272 ready-made pulse and noise sounds** in a set of preset files. Copy them to **GB/Instruments** on the SD card. They are named for how they sound, like KICK, SNARE, HAT, LEAD and PLUK. They are ready to load into any song.

11. Drum kits and samples

The WAV track can play short recorded sounds, like drums. A set of these sounds is called a **kit**. A kit holds up to **15 sounds** and about **2.8 seconds** in total.

Game Boy sound is very rough. PTSDJ changes your sounds into the Game Boy style (4-bit, about 11,000 samples a second). This gives that crunchy retro sound.

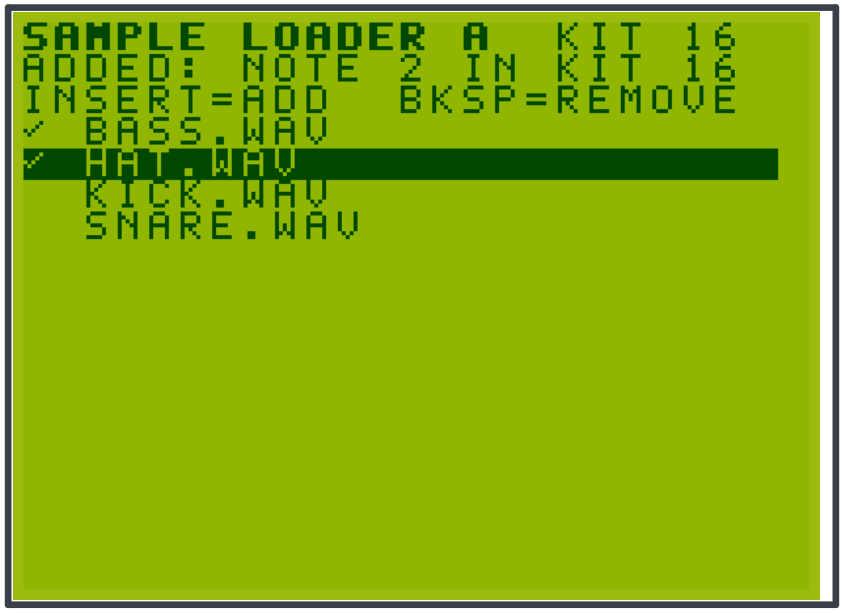
Kit numbers

Each kit has a number (hex). The numbers count from 01, just like on a real Game Boy.

Kits	What they are
01 to 15	Built into the LSDJ ROM. You can hear them, but not change them.
16 to 33	**Yours.** There are 30 free slots. Each Game Boy has its own set.

Making a kit from WAV files

1. Copy WAV files into **GB/Samples** on the SD card. Use plain WAV files (8, 16 or 24 bit). Short sounds (under 2 seconds) work best. Any sample rate is fine.
2. Press **Sample Loader**. The kit number is at the top. Turn the jog wheel to pick a kit between 15 and 32.
3. Pick a file with Up and Down. Press **Play** to hear it as a Game Boy would play it.
4. Press **Insert**. The file is changed and added to the kit. A check mark appears next to it.



The Sample Loader. A check mark means the file is already in the kit.

Key	What it does
Up or Down	Pick a file.
Insert	Add the file to the kit.
Shift + **Insert**	Open it in the Sample Editor first, so you can cut it.
Delete	Take that file's sound out of the kit.
Play	Hear it.
Jog wheel	Pick the kit number.

> **TIP**
>
> Long files are cut to fit the kit. Every change is saved to the SD card at once.

The Sample Editor

Press **Sample Editor** to cut a sound, name it, and add it to a kit. You see the shape of the sound. The two lines show where it starts and ends.



The Sample Editor.

Row	What it does
START and END	Where the sound begins and ends. Turn the jog wheel to move a little. Hold **Insert** and press Up or Down to move a lot.
NAME	Three letters. Press **Insert**. The keyboard starts empty. Pick three letters with **Insert**, then press **Play** for OK. **Delete** cancels and keeps the old name.
KIT	Which kit to add to (16 to 33).
SAMPLES	Shows how many sounds the kit has and how much time is free. Turn the jog wheel to step through the sounds already in the kit so you can change one.
LEVEL	How loud the sound is stored: 100%, 75%, 50% or 25%. Use a lower level for sounds you will layer. It goes back to 100% after each add.
ADD TO KIT	Press **Insert** to add the sound as a new one. The kit is saved at once. When you are on one of the kit's own sounds, this row says REPLACE SAMPLE instead.
DELETE THIS SAMPLE	Throws the sound away. If it is from the kit, the row says DELETE SAMPLE n FROM KIT. It asks first, then erases it and frees its space.
Play	Hears the cut sound.

CAREFUL

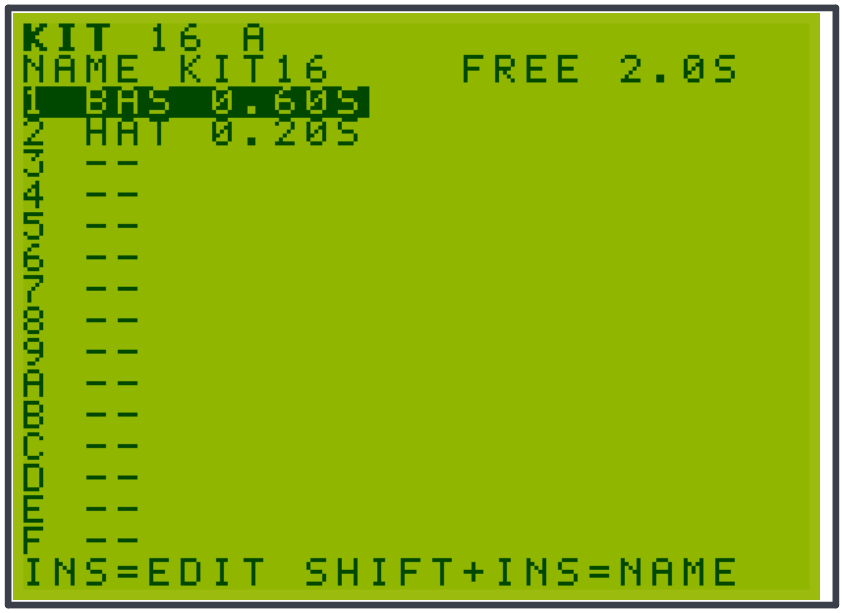
Replacing or deleting a sound that is already in a kit cannot be undone. PTSDJ asks you first.

The red light

While a sound plays on the sample pages, watch the little red light on the screen frame. It **flickers** when the sound is very loud. Loud sounds crackle when you layer them. If it flickers, lower LEVEL until the light stays dark.

The kit page

Press **Sample Playback** to see the sounds in a kit. Use Left and Right at the top to change the kit number.

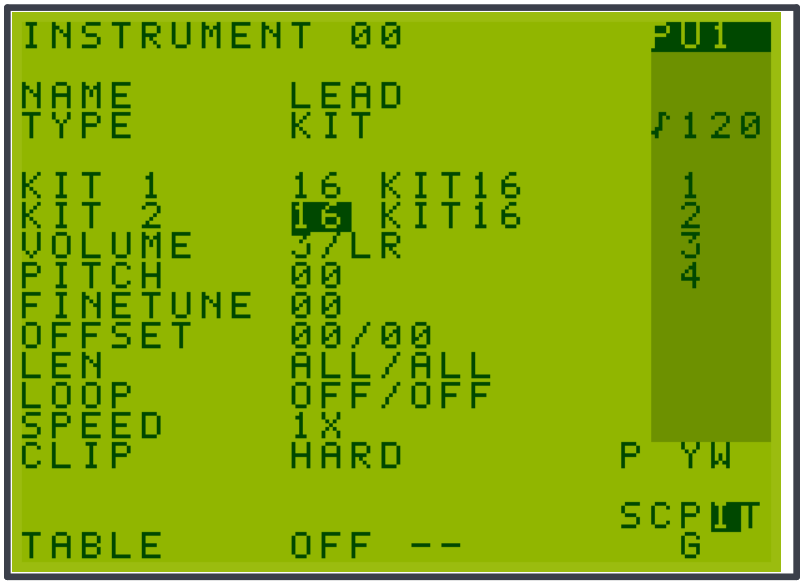


The kit page. Each line is one sound with its length. FREE shows the room left.

Key	What it does
Up or Down	Pick a sound.
Play or a pad	Hear it. Pad 1 plays sound 1, and so on.
Insert	Open the sound in the Sample Editor. It works on any kit, even the ROM kits, so you can copy a sound into your own kit.
Shift + **Insert**	Rename the sound (your kits only). Type three letters, then press **Play**.
Delete	Erase the sound. Press **Insert** to say yes.
NAME row	Press **Insert** to name the kit (up to 8 letters).

Using a kit in a song

1. Go to the Instrument page. Set TYPE to **KIT**.
2. Set **KIT 1** to your kit number (for example 16).
3. Go to the Phrase page. Press **Insert**. The NOTE box shows the **name of a sample**, like KIC or HAT.
4. Hold **Insert** and press Left or Right to pick another sound from the kit.

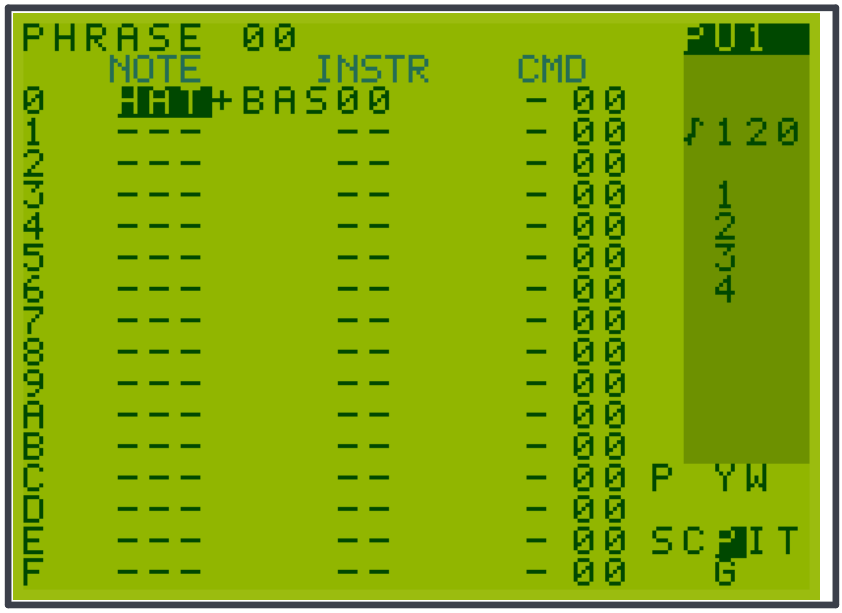


A kit instrument with KIT 1 and KIT 2 both set to kit 16.

Layering two sounds

A step can play a sound from KIT 1 and a sound from KIT 2 **at the same time**. This makes bigger drums. Do it like this:

1. On the Instrument page, set both KIT 1 and KIT 2.
2. On the Phrase page, put in a note. Hold **Insert** and press Left or Right to pick the KIT 1 sound.
3. Hold **Insert** and press Up or Down to add a KIT 2 sound. The step shows both names with a plus, like BAS+HAT.



Layered steps show two names: the KIT 1 sound, a plus, and the KIT 2 sound.

If you press **Insert** with Up or Down on an **empty** step, you get only the KIT 2 sound. The KIT 1 part stays empty (---).

GOOD TO KNOW

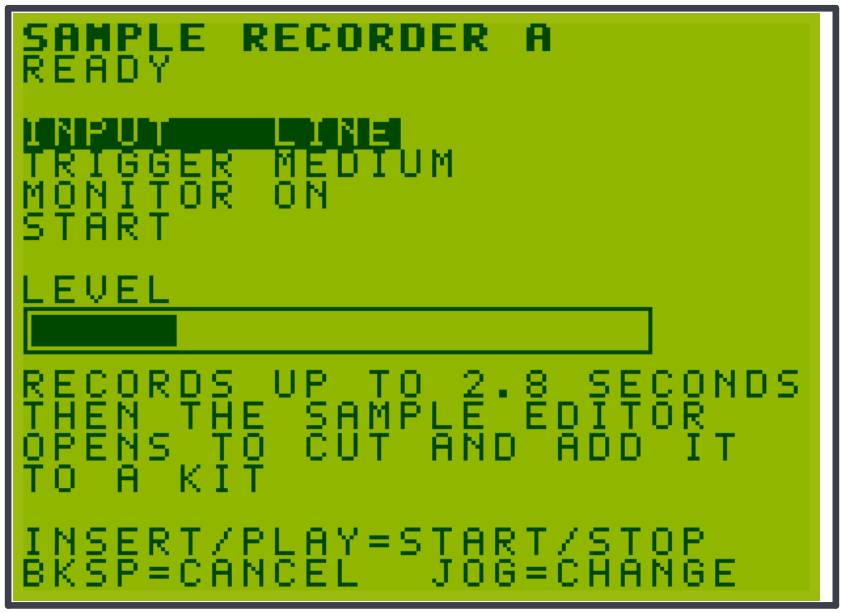
Two sounds are **added** together. If both are loud, the sum is too big and it crackles (the same way real LSDJ does it). Add sounds you plan to layer with LEVEL 50%. Watch the red light in the Sample Editor.

CAREFUL

Only one sound can play on the WAV track at a time. A new note on that track cuts the old one.

Recording your own sound

The Tracker can record from its **line input** or its **microphone** and turn the sound into a kit sample.



The Sample Recorder.

Row	What it does
INPUT	LINE for a cable from another device. MIC LOW or MIC HIGH for a microphone. Turn the jog wheel to pick.
TRIGGER	When recording starts. AT ONCE starts when you press **Insert**. LOW, MEDIUM or HIGH wait until the sound is that loud.
MONITOR	ON lets you hear the input through the Tracker so you can set the level. Turn it OFF with a microphone, or the speakers will squeal.
START / STOP	Press **Insert** or **Play**. It records up to 2.8 seconds, or until you press it again.
LEVEL	A meter showing how loud the input is. Keep it out of the very top.

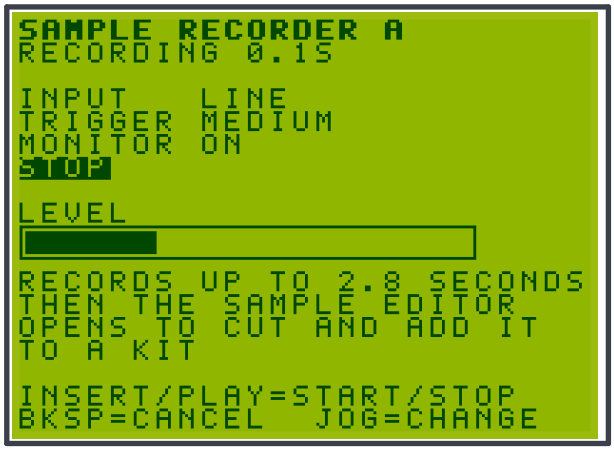
1 Press **Sample Recorder**. Pick the INPUT.

2 Set TRIGGER to MEDIUM so it starts when the sound starts.

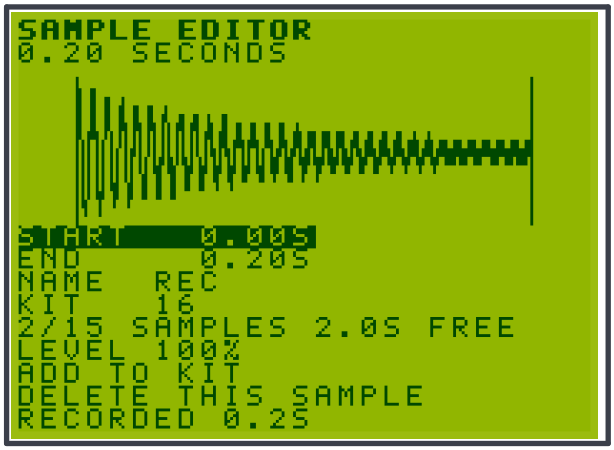
3 Move to START and press **Insert**. The screen says WAITING FOR SOUND.

4 Make the sound. When it stops, the Sample Editor opens with your sound, named REC.

5 Cut it, name it, and add it to a kit.



Recording in progress.



The Sample Editor with the new sound.

TIP

Delete while recording stops and throws the recording away. **Delete** when not recording goes back.

Tips for good drum sounds

- Pick short sounds. A kick or a hat is only a fraction of a second.
- Trim the quiet start and the long tail with START and END.
- Loud, simple sounds work best. Quiet, complex sounds turn to hiss.
- Give each sound a short name so you remember it. KIC, SNR, HAT, CLP.
- A kit has 2.8 seconds in all. Short sounds let you fit more of them.

12. Taking your songs to a real Game Boy

When you finish a song, you can play it on a real Game Boy that runs LSDJ, on a flash cartridge such as an EverDrive, or in an emulator. This chapter shows how.

What you get

On the Project page, move to **EXPORT LSDJ**. Turn the jog wheel to pick **A**, **B** or **A+B**. Press **Insert**. A progress bar runs. The files go in **GB/Export**:

File	What it is
PTSDJ_A.srm	The save file for Game Boy A. It holds your songs.
PTSDJ_A.gb	The LSDJ ROM with Game Boy A's drum kits inside.
PTSDJ_B.srm and PTSDJ_B.gb	The same for Game Boy B.
(song name).lsdsng	A single song in the usual LSDJ song format.

Using them

1. Take the SD card to your computer. Copy **PTSDJ_A.gb** and **PTSDJ_A.srm** from GB/Export.
2. Put both files on your flash cartridge or in your emulator's folder. **Keep the same name** for the two files. The save file goes with the ROM of the same name.
3. Start the ROM. Open your song in LSDJ. Press Start.

> **CAREFUL**
>
> Always use the .srm with the .gb that has the same name. The .gb holds your drum kits. Without it, your kit sounds are missing.

Why a new ROM?

In LSDJ the drum kits are part of the ROM. They are not part of the song. So PTSDJ builds a ROM that has **your kits** in the empty kit slots 16 to 33. It also fixes the ROM's check numbers so flash carts accept it. Only the kit parts and those numbers change. The rest of the ROM is exactly your LSDJ.

Songs that sound a little different

PTSDJ copies LSDJ very closely. But the real Game Boy is the final word. A few small things can sound a little different, like the timing of a vibrato or a sweep. If a sound is not what you expected, trust the Game Boy and adjust the song.

Checklist

- You have your own **GB/LSDJ.GB** on the SD card.
- You saved the song before you exported.
- You copied the .gb and the .srm with the same name.
- Your kit numbers are between 15 and 32.
- On the Game Boy, you loaded the song from the save file.

13. Updating the firmware

You can load a new firmware two ways. Use whichever is easier.

Way 1: from inside PTSDJ

1. Copy the new **.ptf** file into the **Firmware** folder on the SD card.
2. Open the Project page. Hold **Shift**. The ? over F8 turns into **FIRMWARE**.
3. Press F8. A window lists the files in the Firmware folder.
4. Pick one with Up and Down. Press **Insert**. It asks if you are sure. Press **Insert** again to install.
5. The Tracker restarts when it is done.



Hold Shift on the Project page: F8 says FIRMWARE.



The firmware window. The current version is at the top.

> **TIP**
>
> Any other key at the question says no. **Delete** goes back.

Way 2: the restart file

Put the .ptf file and **ot_restore.txt** in the Firmware folder. The text file has one line: the exact name of the .ptf. Restart the Tracker. This works even when PTSDJ is not installed yet.

Going back to the normal Tracker

Pick the stock Polyend .ptf file the same way. Your songs, kits and sounds stay safe on the SD card.

14. When something goes wrong

Find your problem in the left column.

What is wrong	What to try
I hear nothing when I press Play.	Is the track muted? On the Song page, **Shift** + **Insert** mutes and unmutes. Check the instrument: OUTPUT must not be OFF. Check the volume of the instrument and the Tracker's own volume. Is there a note in the phrase and a chain in the song?
The pads make no sound.	PRELISTEN on the Project page may be OFF. Turn it ON. Or turn **Rec** on and use the pads on the Phrase page.
The Synth page, the Wave page or the kits do not work.	They need **GB/LSDJ.GB** on the SD card. Check the name and the folder.
My kit sounds are missing on the Game Boy.	Use the **.gb** that PTSDJ exported, together with the .srm of the same name. Kit numbers must be 16 to 33. A song that uses a kit your ROM does not have plays noise.
I cannot change kits 01 to 15.	They are the ROM's own kits. Copy a sound out of one (Sample Playback, **Insert**) into your own kit 16 to 33.
Layered drums crackle.	The two sounds are too loud together. Lower LEVEL in the Sample Editor to 50% when you add them. Watch the red light. Songs made with an early version of PTSDJ are repaired when you save or export.
Sample times on the screen show only an S.	This was fixed in version 0.8.5. Use version 1.0 or newer.
SAVE says BANK FULL.	The save file has 32 songs, or it is out of room. Erase an old song in the song list (**Shift** + **Delete**), or run CLEAN SONG DATA.
SAVE FAILED or CAN'T FLASH.	Check the SD card. It must not be locked, full or loose. Try saving again.
The recorder hears nothing.	Check the cable and the INPUT row. Try TRIGGER set to AT ONCE. You see NOTHING HEARD if the sound is too quiet.
The recorder squeals.	MONITOR is ON with a microphone and speakers. Turn MONITOR OFF.
The firmware update does not start.	The ot_restore.txt must hold only the file name, with no extra lines or spaces. The .ptf must be in the Firmware folder.
My Mac left strange files on the card.	Files that start with ._ are hidden by PTSDJ. You can ignore them.
It sounds a little different from LSDJ on the Game Boy.	Some small timing details differ. The real Game Boy is the final word. Chapter 12 explains.
I want to start over.	Use NEW SONG. Or erase songs in the song list. Or go back to the stock firmware (Chapter 13).

15. Words to know

Word	What it means
BKSP	Backspace. The **Delete** key. Clears things or goes back.
Chain	A list of up to 16 phrases.
Command	A letter and two digits that does a special trick on one step.
Cursor	The bright box that shows where you are.
Envelope	How the loudness of a note changes over time.
Export	Make files you can use somewhere else, like on a Game Boy.
Groove	A list that says how many ticks each step lasts.
Hex	Counting with the letters A to F after 9.
Instrument	A sound you can play.
Kit	A set of up to 15 short recorded sounds.
Layer	Play two sounds at the same time.
LSDJ	Little Sound DJ, a music program for the Game Boy.
Octave	Twelve half steps. The same note, but higher or lower.
Phrase	Sixteen steps. Where the notes live.
ROM	The program file for a Game Boy cartridge.
Sample	A short recorded sound.
Song	The big plan that tells each track which chain to play.
Step	One of the 16 beats in a phrase.
SRM	The file type for a Game Boy save file.
Table	A tiny pattern that changes a sound while it plays.
Tick	One tiny tap of the music clock. A step lasts some ticks.
Track	One of the four sound makers: PU1, PU2, WAV, NOI.
Transpose	Move notes higher or lower.

Appendix A: Every key

Keys	Do this
Insert	Put in a value. Open the name keyboard. Press the main button.
Insert + Left or Right	Change by 1.
Insert + Up or Down	Change by 16, or by an octave on notes.
Jog wheel	Change a number. Pick a kit. Pick a sound. Same as **Insert** + Left or Right.
Delete	Clear a box. Go back. Say no.
Delete + Left or Right	Pick another chain, instrument, table, groove or wave.
Delete + Up or Down	Pick another phrase (Phrase page).
Shift + Left, Right, Up, Down	Walk to the next page.
Shift + **Insert** + arrows	Pick many boxes.
Copy	Copy the picked boxes.
Shift + **Copy**	Paste.
Play	Start or stop. Plays just a chain, phrase or table on those pages.
Shift + **Play**	Play the whole song from the Chain, Phrase and Table pages.
Rec	Recording from the pads.
Shift + **Insert** (Song page)	Mute or unmute a track.
Shift + **Delete** (Song page)	Mute all four tracks.
Shift + **Delete** (song list)	Erase a song from the save file.
Shift + **Song**	Swap song A and song B.
Shift + **Instrument**	Open the instrument library.
Shift + **Note** (Phrase page)	The command cheat sheet.
Shift + **Instrument Parameters** (Table page)	The command cheat sheet.
Shift + a pad row	Change the octave of the pads.
Shift + **Insert** (Sample Loader)	Open the file in the Sample Editor.
Shift + **Insert** (kit page)	Rename a sound.
Shift + F8 (Project page)	Open the firmware window.
Hold F8	A pop-up about the page you are on.

Keys	Do this
F1 and F2 (while the pop-up shows)	Turn its pages back and forward.
Master	Show both Game Boys side by side.

Appendix B: Limits

Thing	How many
Tracks	4 per Game Boy: PU1, PU2, WAV, NOI. 8 in all.
Song rows	256
Chains	128 (00 to 7F), each with up to 16 phrases
Phrases	255 (00 to FE), each with 16 steps
Instruments	64 (00 to 3F)
Tables	32 (00 to 1F), each with 16 rows
Grooves	32 (00 to 1F), each with 16 steps
Synths	16 (0 to F)
Tempo	28 to FF (hex)
Songs in one save file	32
Your own kits	30 (kits 16 to 33) per Game Boy
Sounds in a kit	15, and about 2.8 seconds in all
Recording	Up to 2.8 seconds at a time
Instruments in the library	Up to 400 files

Appendix C: About PTSDJ

PTSDJ is a personal project that makes the Polyend Tracker work like a Game Boy music program. It is for your own use.

- PTSDJ is built on the open-source Polyend Tracker firmware, which allows **non-commercial** use only.
- LSDJ (Little Sound DJ) was made by Johan Kotlinski. PTSDJ does not include LSDJ. You need your own copy of the ROM.
- PTSDJ is not made by or connected to Polyend or to the makers of LSDJ.
- Normal Polyend Tracker projects are never touched by PTSDJ. You can always go back to the stock firmware.

Version

This guide matches PTSDJ version 1.0.0.